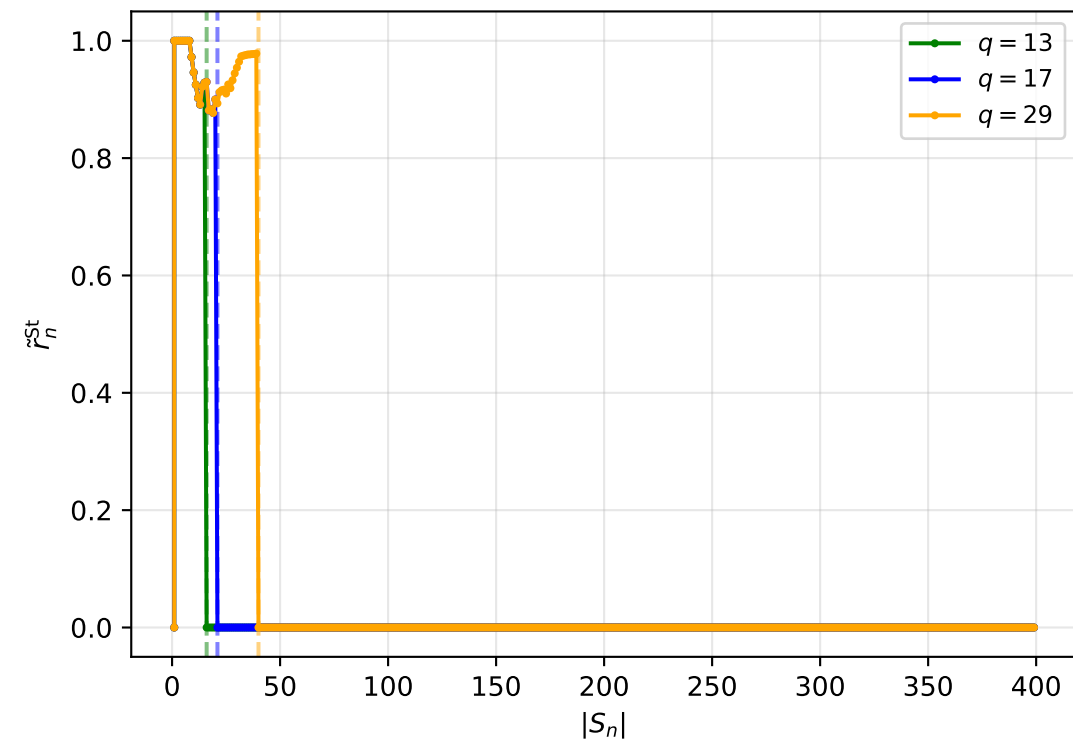
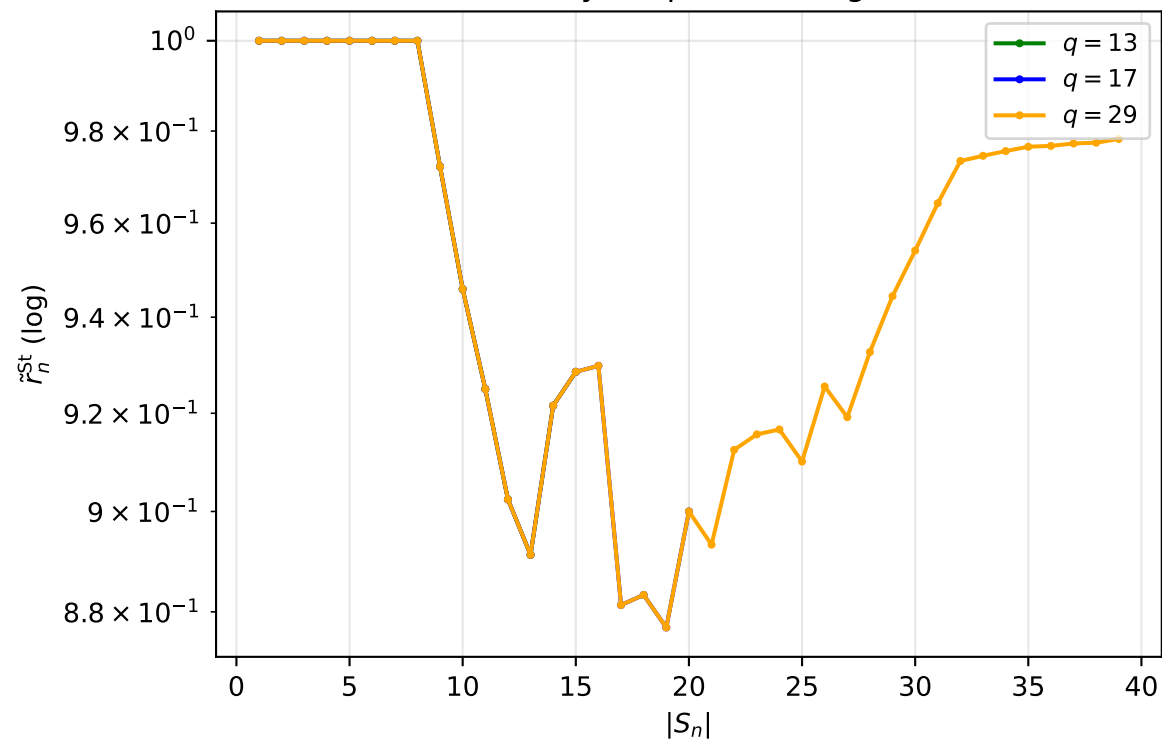


Steinberg $\tilde{\sigma}_u \odot \tilde{\sigma}_s$ pre-saturation law -- $X^{5,q}$, $q \in 13, 17, 29$

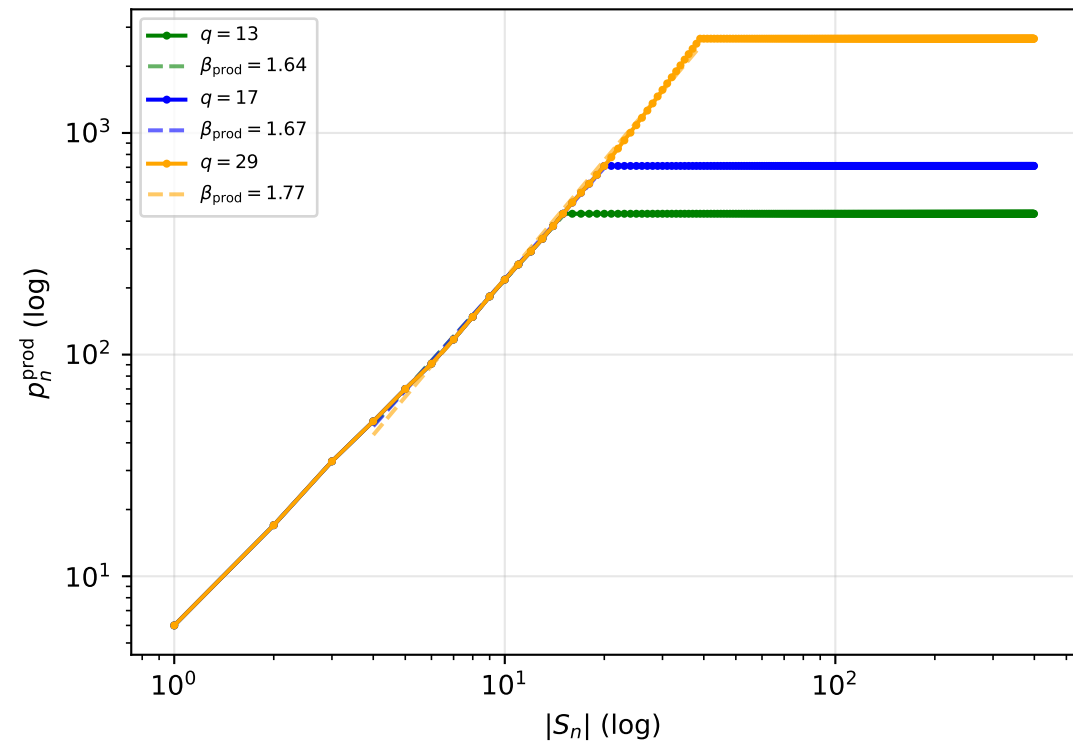
Admissible frontier fraction (linear)



Decay shape (semi-log)



Cumulative admissible front (log-log)



Summary

PARAMETER STABILITY

Saturation thresholds vs q :

$q=13$: $|S^*| = 16$, $|S^*|/|G| = 0.00366$

$q=17$: $|S^*| = 21$, $|S^*|/|G| = 0.00054$

$q=29$: $|S^*| = 40$, $|S^*|/|G| = 0.00023$

$|S^*| \sim q^{1.15}$

β_{prod} stability:

$q=13$: $\beta_{\text{prod}} = 1.6438$

$q=17$: $\beta_{\text{prod}} = 1.6682$

$q=29$: $\beta_{\text{prod}} = 1.7726$

mean: 1.6949, std: 0.0559

Note: $\beta_{\text{prod}} \approx 1.7$ is outside

$\beta^* \in (0.09, 0.13)$; governed by

$O(q)$ pre-saturation window only.